

Modeling and Simulation — Lesson 2

Feedback in the Nature

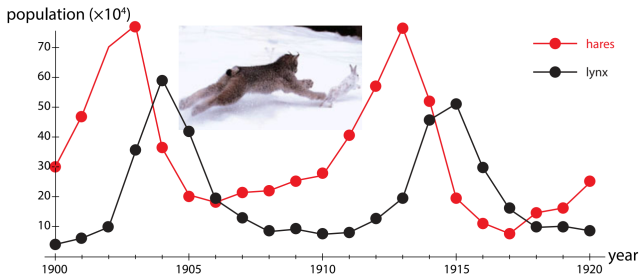
Dynamics of Predator-Prey Interactions

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Predator-Prey



- ▶ The Lotka-Volterra equations, also known as the predator-prey equations, model the dynamics of biological systems in which two species interact, one as a **predator** and the other as **prey**.
- ▶ They were proposed independently by Alfred J. Lotka in 1925 and Vito Volterra in 1926.
- ▶ Model is characterized by **oscillations** in the population size of both predator and prey,
- ▶ Peak of the predator's oscillation lags behind the peak of the prey's oscillation.

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- ▶ predators can consume any quantities of prey (unrealistic for large quantities)
- ▶ both populations are moving randomly through a homogeneous, static, environment (may be approximately true for some environments)

The Mathematical Model

The Lotka-Volterra equations are a pair of first-order, nonlinear, differential equations given by:

$$\frac{dN}{dt} = rN - \beta NP \quad (1)$$

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- ▶ β attack efficiency of the predator

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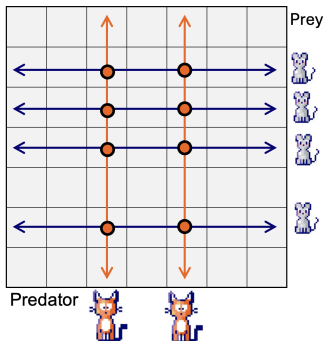
The reduction in prey population due to predators is captured by the expression:

$$-\beta NP \quad (5)$$

Here, NP is proportional to the number of encounters. An increased number of predators, an increased prey population, and greater searching efficiency of the predators would all lead to a higher removal rate of prey. β is attack efficiency of the predator. Higher β means higher probability of prey being killed.

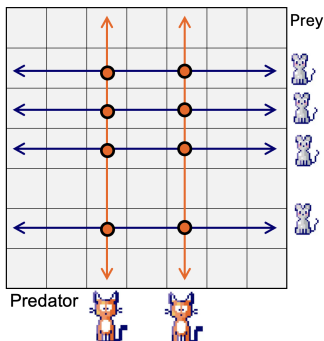
Predator-prey encounters

Picture below illustrates the concept of predators-prey encounters as $N \cdot P$ product.



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The concept would not work for predators hunting in packs or for prey staying in herds.

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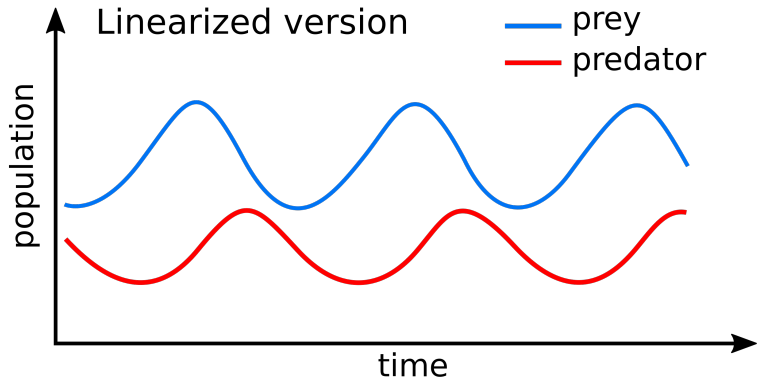
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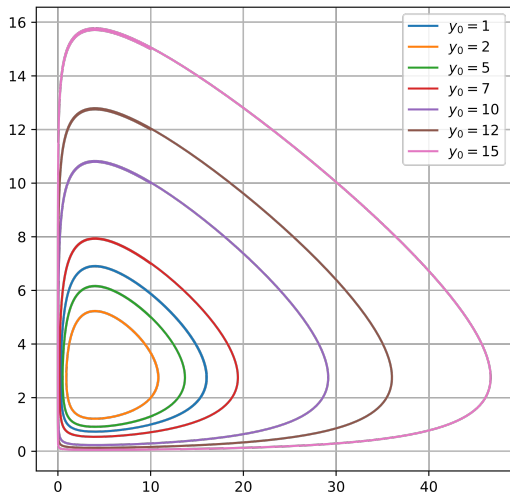
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The growth of predator population growth by eating of prey

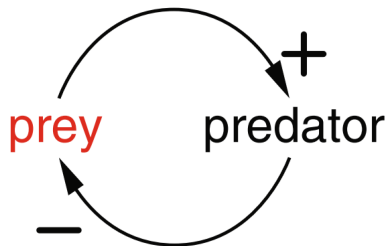
$$+m\beta NP \quad (8)$$

Here we see again βNP . Proportionality constant m tells how many prey individuals are required for increase of predator population by one.





Concept of feedback



1

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- ▶ Visually, this is represented as a grid of arrows. The **direction** of the arrow shows where the system is heading, and its **length** (magnitude) shows how fast it is changing.
- ▶ Vector fields allow us to understand the qualitative, long-term behavior of a system without needing to analytically solve the complex differential equations.

Vector Field in Predator-Prey Phase Space

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- ▶ The arrows in this specific vector field indicate how both populations will shift from that exact moment.
- ▶ By following the continuous flow of these arrows, we can trace the closed, periodic trajectories of the populations over time.

Calculating the Vector Field: Examples

- ▶ Let's calculate the instantaneous rate of change $\vec{v} = \left(\frac{dN}{dt}, \frac{dP}{dt}\right)$ using a specific set of parameters:
- ▶ **Assume:** $r = 2$ (prey growth), $\beta = 0.1$ (attack rate), $m = 0.5$ (conversion), and $d = 1$ (predator death).
- ▶ Our rate equations become:

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- ▶ **Point 1: (10, 10) — Low prey, low predators**

$$\vec{v} = (2(10) - 0.1(10 \cdot 10), 0.05(10 \cdot 10) - 10) = (10, -5)$$

Result: Prey population grows (+10), predators starve (-5).

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- ▶ **Point 2: (40, 10) — Abundant prey, low predators**

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Result: Both populations are booming.

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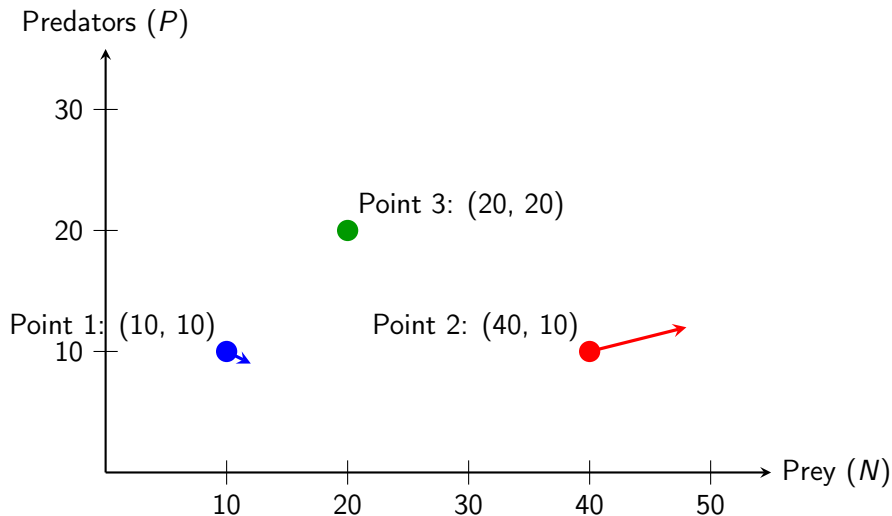
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- ▶ **Point 3: (20, 20) — The Coexistence Point**

$$\vec{v} = (2(20) - 0.1(20 \cdot 20), 0.05(20 \cdot 20) - 20) = (0, 0)$$

Result: The system is perfectly balanced; no change.

Visualizing the Vector Field



Equilibrium Points in the Vector Field

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- ▶ The vector field arrows orbit continuously around the coexistence equilibrium point, mathematically illustrating the perpetual lagging and catching-up dynamics of the two species.

Biological Implications

- ▶ The model predicts oscillations in the populations of predator and prey.
- ▶ The fluctuation of the two species is periodic,
- ▶ The amplitude and period of these oscillations depend on the initial conditions and the parameters of the model.
- ▶ The *average* numbers of the two species are constant
- ▶ If we destroy individuals of both species proportionately to their number, the average number of individuals of the prey grows and the average number of predators diminishes.

Conclusion

- ▶ The Lotka-Volterra predator-prey model provides a framework for understanding the dynamics of biological systems with predator-prey interactions.
- ▶ Despite its simplicity, it offers valuable insights into the oscillatory nature of ecological populations and has motivated further research into more complex and realistic models.

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- ▶ Substituting our Lotka-Volterra equations, we establish a system of two algebraic equations:

$$rN - \beta NP = 0 \quad (13)$$

$$m\beta NP - dP = 0 \quad (14)$$

Calculating Equilibrium Points: Step 2

- ▶ Next, we factor N and P out of both equations to simplify solving them:

$$N(r - \beta P) = 0 \quad (15)$$

$$P(m\beta N - d) = 0 \quad (16)$$

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- ▶ **Case 1: The Extinction Equilibrium**
- ▶ From the first equation, one obvious solution is $N = 0$.
- ▶ Substituting $N = 0$ into the second equation gives $P(-d) = 0$. Since the death rate $d > 0$, this means P must be 0.
- ▶ This gives us our first equilibrium point at the origin: $(0, 0)$.

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- ▶ Solving these for P and N respectively:

$$P = \frac{r}{\beta} \quad \text{and} \quad N = \frac{d}{m\beta} \quad (19)$$

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- ▶ This yields our second, non-trivial equilibrium point: $\left(\frac{d}{m\beta}, \frac{r}{\beta}\right)$.

Modeling Approaches

Modeling Approach 1: The Euler Method

- ▶ The Euler method is the simplest numerical procedure for solving ordinary differential equations (ODEs).
- ▶ **How it works:** We break time into discrete, tiny steps (Δt). At each step, we calculate the current rate of change and assume it remains constant for that brief interval.
- ▶ **Discrete Equations:**

$$N_{t+\Delta t} = N_t + \Delta t(rN_t - \beta N_t P_t) \quad (20)$$

$$P_{t+\Delta t} = P_t + \Delta t(m\beta N_t P_t - dP_t) \quad (21)$$

- ▶ **Pros:** Transparent; excellent for learning how continuous equations translate to discrete steps.
- ▶ **Cons:** Prone to accumulating errors. If Δt is too large, the inherent closed loops artificially spiral outwards.

Euler Method in Python

Demonstrating the explicit loop, mapping directly to the discrete equations:

```
1 import numpy as np
2
3 r, beta, m, d = 1.0, 0.1, 0.05, 0.5
4 dt = 0.01 # Small time step is crucial for stability
5 steps = 2000
6
7 N, P = np.zeros(steps), np.zeros(steps)
8 N[0], P[0] = 40, 9 # Initial conditions
9
10 for i in range(1, steps):
11     dN_dt = r * N[i-1] - beta * N[i-1] * P[i-1]
12     dP_dt = m * beta * N[i-1] * P[i-1] - d * P[i-1]
13
14     N[i] = N[i-1] + dN_dt * dt
15     P[i] = P[i-1] + dP_dt * dt
```

Modeling Approach 2: ODE Solvers

- ▶ For practical applications and research, manual loops are often insufficient. We use dedicated mathematical solvers, such as SciPy's `solve_ivp`.
- ▶ **How it works:** These solvers act as a "black box" utilizing advanced algorithms (like Runge-Kutta). You provide the differential equations, time span, and initial conditions.
- ▶ **Adaptive Step Size:** The solver automatically shrinks Δt when populations change rapidly (for accuracy) and expands it when stable (to save computing power).
- ▶ **Pros:** Highly accurate, stable, and correctly handles the periodic nature of predator-prey dynamics without artificial spiraling.
- ▶ **Cons:** Less transparent for beginners; the integration loop is hidden behind the scenes.

SciPy Solver in Python

Define the system as a function, allowing the solver to handle integration:

```
1 from scipy.integrate import solve_ivp
2
3 def lotka_volterra(t, state, r, beta, m, d):
4     N, P = state
5     dN_dt = r * N - beta * N * P
6     dP_dt = m * beta * N * P - d * P
7     return [dN_dt, dP_dt]
8
9 params = (1.0, 0.1, 0.05, 0.5)
10 initial_state = [40, 9]
11 t_span = (0, 20)
12
13 # 'RK45' is the default Runge-Kutta method
14 solution = solve_ivp(fun=lotka_volterra, t_span=t_span,
15                       y0=initial_state, args=params)
16
17 time = solution.t
18 prey, predators = solution.y[0], solution.y[1]
```